

Compute Economics — Executive Brief

Integrating five models into a single compute-margin view — the cross-functional decision artifact.

Synthetic portfolio · R. A. Dapaah · github.com/dapaah/compute-economics-portfolio

FIVE HEADLINE INSIGHTS

- 1 Cost per token is a throughput problem.** ProductC runs ~17% above blended cost per 1K output tokens — driven by decode-bound output intensity, not volume. Cost is ~75% compute.
- 2 Batching is the direct cost lever, bounded by latency.** Raising the interactive batch from ~3 to ~14 cuts cost per inference ~39% (+72 ms); the margin-optimal batch under a 150 ms SLO is 16.
- 3 Caching is the highest-quality lever.** Each +10 points of cache-hit rate cuts effective cost per inference ~11.8% — ≈ \$2.3M / year at 1B inferences/day — with no latency penalty.
- 4 Demand sets the capital ask.** Usage compounds ~2.4× / year; the fleet scales from ~1,260 to ~1,985 GPUs (base) within a year — ≈ \$22M of take-or-pay capital (band ~1,770–2,200).
- 5 Utilisation, not mix, is the remaining margin lever.** A GPU-second earns ~47% more margin in interactive than batch, but capacity is already ~77% interactive; the live lever is lifting utilisation from ~76% toward the ~85% ceiling.

THE INTEGRATED STORY

cost → cache → demand → allocation → margin. Margin on a compute fleet is won by raising effective throughput per GPU-second. Cost originates in the decode phase; two levers reduce it — caching first, because it carries no latency penalty, then batching up to the SLO. Demand compounding ~2.4×/year sets the scale of the capital bet, and because capacity already favours the high-margin interactive tier, the binding constraint is utilisation rather than further mix shifts.

DECISIONS THIS INFORMS

- **Pricing / routing:** reprice or route output-heavy ProductC to higher-throughput GPUs (H200/B200).
- **Engineering ROI:** fund cache-hit improvements — ~\$2.3M per +10 points clears most engineering costs.
- **Capital:** stage GPU procurement against the ~430-GPU downside/upside band, not a single commit.
- **Operations:** drive utilisation toward ~85% under the latency SLO as the primary remaining margin lever.

Illustrative figures from a synthetic 120,000-event inference dataset; the framework scales linearly to production volume. Full data pipeline and five model notebooks: github.com/dapaah/compute-economics-portfolio